# 13.8 Classifying Quadrilaterals – Part 1

## Lesson Introduction

| | | | |
|---|---|---|---|
| Benchmark | MA.912.GR.3.2 Given a mathematical context, use coordinate geometry to classify or justify definitions, properties, and theorems involving circles, triangles, or quadrilaterals. | Learning Targets | • I can use coordinate geometry to classify quadrilaterals based on their properties in a mathematical or real-world context.<br>• I can draw a quadrilateral on a coordinate plane using given definitions, properties, or theorems. |

| | Building On | Working Toward |
|---|---|---|
| Benchmark Coherence | N/A | N/A |

| | |
|---|---|
| Essential Questions | • How can quadrilaterals be classified according to their properties?<br>• What justifications can be given to prove that a shape is a certain type of quadrilateral?<br>• How do you find the lengths of sides and diagonals of quadrilaterals? |
| Lesson Narrative | Throughout Lesson 13.8, students work on classifying quadrilaterals on a coordinate plane. This lesson guides students to use the Pythagorean Theorem to find the lengths of the sides and diagonals of quadrilaterals. Students will use slope to determine if lines are parallel, perpendicular, or neither. Students will then classify the quadrilateral using the specific properties of each type of quadrilateral. |

| | | | | |
|---|---|---|---|---|
| Component Summary | 13.8.1 Warm-Up (~5 mins) | Comparison of measurements *(SE p. 327)* | 13.8.4 Guided Practice (15 mins) | Drawing, Classifying, and Justifying Quadrilaterals *(SE p. 331)* |
| | 13.8.2 Exploration (10 mins) | Properties of Quadrilaterals *(SE p. 328)* | 13.8.5 Your Turn! (5 mins) | Classifying quadrilaterals Independently *(SE p. 332)* |
| | 13.8.3 Exploration (10 mins) | Classifying Quadrilaterals *(SE p. 329)* | 13.8.6 Wrap-Up (~5 mins) | Error Analysis *(SE p. 333)* |

| | Glossary | Materials | Additional Preparation |
|---|---|---|---|
| Resources | <u>Key Terms:</u> N/A<br><br><u>Additional Terms:</u><br>• Quadrilateral<br>• Vertices | • Calculators<br>• (optional) Graphing utility | • N/A |

Teacher Tips

| Common Misconceptions | Things to Consider |
| --- | --- |
| • Students may make an error when using the Pythagorean Theorem or Distance Formula to find the lengths of sides or diagonals of a quadrilateral.<br>• Students may forget the properties for each quadrilateral. | • 13.8.2 Exploration spirals content from Unit 8. Consider using information from assessments and observational data to understand what misconceptions from unit 8 may perpetuate in this unit. Consider how to address these misconceptions, and where small group instruction may be helpful. |

| Literacy and Language Routines | Mathematical Thinking and Reasoning (MTR) Standard |
| --- | --- |
| **Discussion Supports**<br>There are several places within the lesson that students are asked to discuss with their partners. While going through the Guided Practice, consider having students turn to their partner to discuss what they just learned. This opens the lesson to have students justify their work with the discussion. Use this with 13.8.2 Exploration, questions 2-3, and 13.8.4 Guided Practice, question 4. Listen to students' conversations and consider asking higher order questions to get the students thinking about their justifications. | **MA.K12.MTR.1.1: Participate in Effortful Learning**<br>Students will be working on classifying quadrilaterals in the coordinate plane. As students explore, they can ask their partners clarifying questions to deepen their understanding. The Explorations and Guided Practice provide help and support for students with the new concept. In 13.8.5 Your Turn! students participate in individual, effortful learning, while working through complex problems. |

Differentiate Instruction

Visuals of all types of quadrilaterals can help a learner see what properties they are applying to what shape, to classify it correctly. The use of GeoGebra, as a class or on student devices, can help the students to see the drawings and be able to identify some of the properties applied to the shapes, to help with classification of all types of quadrilaterals. Discussions throughout the lesson will assist students who benefit from auditory entry points.

Lesson Notes

# 13.8.1 Warm-Up: Would You Rather?

***Component Overview:*** Students will identify which amount of gold they would rather have, where they have the choice between a 14 m cube of gold or two cubes — a 12 m cube and the other an 8 m cube — combined.

## 13.8.1 Warm-Up: Would You Rather?

Gold is worth a significant amount of money. Its value changes with the market. For example, on January 28, 2021 the value of 1 ounce of gold was nearly \$2000.

1. Would you rather have one cube made of solid gold measuring 14 meters (m) on each side, or two cubes made of solid gold, one measuring 12 m on each side and one measuring 8 m on each side?


*Answers will vary.*

2. Explain your choice.
*Answers will vary. I would take the 14-meter square because it is the biggest square. I would take the two 12 meter and 8-meter squares because they have more surface area.*

Ask the following question to connect the Warm-Up to the lesson:

- To compare the weight of the gold, is it better to find the surface area or the volume?

# 13.8.2 Exploration: Classifying Quadrilaterals Using Properties

***Component Overview:*** *The Exploration has students using a word bank to classify quadrilaterals by specific properties. Students will discuss with their partner about certain properties to see if it can be used to be classified as a quadrilateral or not.*

## 13.8.2 Exploration: Classifying Quadrilaterals Using Properties

1. Complete the table by filling in the blanks using the word bank. Words may be used more than once, or they may not be used.

| Word Bank | |
|---|---|
| bisect(s) | congruent |
| parallel | perpendicular |

| Classification | Property |
|---|---|
| Parallelogram | Opposite sides are ___*parallel*___ and ___*congruent*___. |
| Parallelogram | Diagonals ___*bisect*___ each other. |
| Rectangle | Diagonals are ___*congruent*___. |
| Rhombus | Diagonals are ___*perpendicular*___. |
| Square | Diagonals are ___*perpendicular*___ and ___*congruent*___. |
| Kite | Exactly one diagonal ___*bisects*___ the other, and diagonals are ___*perpendicular*___. |
| Isosceles Trapezoid | Diagonals are ___*congruent*___ and at least one pair of sides are ___*parallel*___. |

2. Discuss with your partner if knowing two consecutive sides are congruent is sufficient to classify a quadrilateral. Summarize your discussion and justify your decision.
*Answers will vary.*
*No. There is more than one type of quadrilateral that has congruent consecutive sides. For example, this is the case for both kites and squares.*

*Allow students to review their notes and Unit 8 to recall the properties. Research shows that this type of spaced recall helps students solidify the information.*

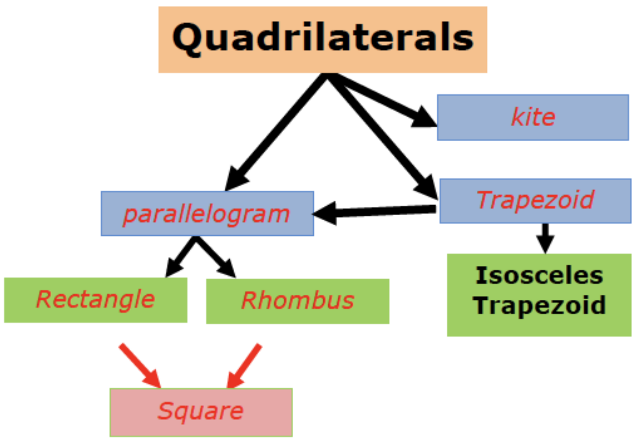
*While students are working, consider collecting observational data, using a checklist of the quadrilaterals. Specifically, focus on what attributes students struggled remembering or finding. Use the observations from 13.8.2 and 13.8.3 to inform what differentiation or small-group instruction may be helpful.*

*Consider having drawings of the quadrilaterals to help in classifying them, according to specific properties.*

# 13.8.2 Exploration: Classifying Quadrilaterals Using Properties (continued)

3. Discuss with your partner if knowing a pair of opposite sides are congruent is sufficient to classify the quadrilateral. Summarize your discussion and justify your decision.
   *Answers will vary.*
   *No. Many types of quadrilaterals have opposite sides that are congruent. Rectangles, squares, and rhombi are a few examples.*

4. Complete the diagram using the different types of quadrilaterals. Each blue box is a subcategory of quadrilaterals. Each green box is a subcategory of the blue box that it is connected to.




5. Which quadrilateral was not included in the diagram?
   *Square*

6. Draw a box and any necessary arrows to include the missing quadrilateral in the diagram.
   *Shown above.*

*Consider asking students who finish early to think of other ways to display the relationships and hierarchy of quadrilaterals.*

*It might be beneficial to have students draw each quadrilateral and label the properties next to the category names to help visual learners know what each quadrilateral looks like.*

*After students complete the diagram, consider putting the diagram on display in the classroom for students to use as a reference.*

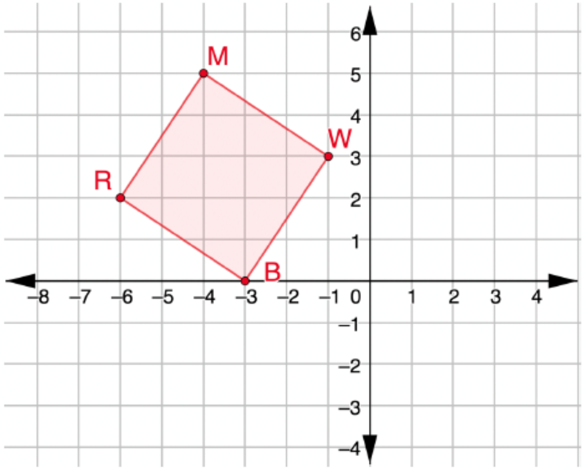
# 13.8.3 Exploration: Classifying Quadrilaterals

***Component Overview:*** *In this Exploration, students will use the properties of quadrilaterals to classify the quadrilateral given.*

## 13.8.3 Exploration: Classifying Quadrilaterals

1. $RMWB$ has vertices at $R(-6,2)$, $M(-4,5)$, $W(-1,3)$, and $B(-3,0)$.

   a. Graph $RMWB$ on the coordinate plane.




   b. Complete the table.

| Line Segment | Length |
|---|---|
| $\overline{RM}$ | $\sqrt{13}$ |
| $\overline{MW}$ | $\sqrt{13}$ |
| $\overline{WB}$ | $\sqrt{13}$ |
| $\overline{BR}$ | $\sqrt{13}$ |

   c. What do you notice about the side lengths of $RMWB$?
   *All the sides lengths are equal.*

*Monitor while students are working to determine if there are any questions that need to be debriefed before moving to 13.8.4 Guided Practice. This could include highlighting student voices to share rich discussions or comments overheard while you circulated, or to address common misconceptions that were evident among the majority of the class.*

*This Exploration is scaffolded to allow students to engage with it with their partner, rather than being guided through by the teacher. Additional scaffolding questions are provided, if needed, but this should be a primarily student-directed Exploration.*

## 13.8.3 Exploration: Classifying Quadrilaterals (continued)

d. Select all of the classifications that apply to $RMWB$.

- ☐ Isosceles trapezoid
- ☐ Kite
- ☐ Parallelogram
- ☐ Rectangle
- ☐ Rhombus
- ☐ Square
- ☐ Trapezoid

*Consider using this question to revisit the formal definition of a trapezoid as a quadrilateral with at least one pair of parallel sides. Use guiding questions such as, "Which quadrilaterals can also be classified as trapezoids?", (i.e., square, rectangle, rhombus, parallelogram).*

e. Complete the table.

| Diagonal Line Segment | Length |
|---|---|
| $\overline{RW}$ | $\sqrt{26}$ |
| $\overline{MB}$ | $\sqrt{26}$ |

- *How can you determine the length of the line segment?*
- *What are the properties used to help in the classification of the quadrilateral?*
- *What information is best used in classifications with side measurements and the diagonal?*

f. What do you notice about the diagonal lengths of $RMWB$?

*The diagonals are congruent.*

g. Complete the statement.

Since $RMWB$ ○ has ○ does not have congruent diagonals, $RMWB$ is classified as a(n) *square or rectangle*.

2. $NTCV$ has vertices at $N(2,3)$, $T(3,5)$, $C(6,6)$, and $V(4,2)$.

a. Complete the table.

| Line Segment | Length |
|---|---|
| $\overline{NT}$ | $\sqrt{5}$ |
| $\overline{TC}$ | $\sqrt{10}$ |
| $\overline{CV}$ | $2\sqrt{5}$ |
| $\overline{VN}$ | $\sqrt{5}$ |

# 13.8.3 Exploration: Classifying Quadrilaterals (continued)

b. What do you notice about the side lengths of $NTCV$?
*Two adjacent sides are the same measurements.*
$NT = VN$

c. Complete the table.

| Line Segment | Slope |
|---|---|
| $\overline{NT}$ | $2$ |
| $\overline{TC}$ | $\frac{1}{3}$ |
| $\overline{CV}$ | $2$ |
| $\overline{VN}$ | $-\frac{1}{2}$ |

d. What do you notice about the slopes of $NTCV$?
*Answers will vary. Quadrilateral $NTCV$ has one set of parallel sides and two set of perpendicular sides.*

e. Select all of the classifications that apply to $NTCV$.

- ☐ Isosceles trapezoid
- ☐ Kite
- ☐ Parallelogram
- ☐ Rectangle
- ☐ Rhombus
- ☐ Square
- ☐ Trapezoid

- *What information do the slopes of the line segments tell us about the quadrilateral?*
- *Why do you need to find the lengths and slopes of the sides for quadrilateral $NTCV$, whereas with quadrilateral $RMWB$, you needed side lengths and diagonals?*
- *Do you need to find the diagonals of quadrilateral $NTCV$? Why or why not?*

# 13.8.4 Guided Practice: Classifying Quadrilaterals

***Component Overview:*** *Students work independently to practice justifying quadrilaterals as specific shapes, and how to use the coordinate plane to help in the justification.*

## 13.8.4 Guided Practice: Classifying Quadrilaterals

1. $GSPD$ has vertices at $G(-2,-3)$, $S(3,-2)$, $P(5,-4)$, and $D(0,-5)$.

   Justify that $GSPD$ is a parallelogram but not a rectangle. Show your work.

   $1^2+5^2=GS^2$
   $\sqrt{26}=GS$

   $2^2+2^2=SP^2$
   $2\sqrt{2}=SP$

   $1^2+5^2=DP^2$
   $\sqrt{26}=DP$

   $2^2+2^2=GD^2$
   $2\sqrt{2}=GD$

   *Quadrilateral $GSPD$ is a parallelogram because opposite sides are congruent.*

   $GP=\sqrt{(5+2)^2+(-4+3)^2}$
   $GP=\sqrt{50}$

   $DS=\sqrt{(0-3)^2+(-5+2)^2}$
   $DS=\sqrt{18}$

   *Since the diagonals are not congruent, the parallelogram is not a rectangle.*

2. $HMTK$ has vertices at $H(-2,4)$, $M(-2,0)$, $T(-6,-1)$, and $K(-6,5)$. Classify $HMTK$ and justify your classification using mathematics.

   *$\overline{HM}$ and $\overline{KT}$ both have undefined slopes thus making them parallel. $\overline{KH}$ has a slope of $-\frac{1}{4}$ and $\overline{TM}$ has a slope of $\frac{1}{4}$, making them not parallel.*

   $KH=\sqrt{1^2+4^2}=\sqrt{17}$
   $TM=\sqrt{1^2+4^2}=\sqrt{17}$

   *Since the non-parallel sides are equal in length, the quadrilateral is an isosceles trapezoid.*

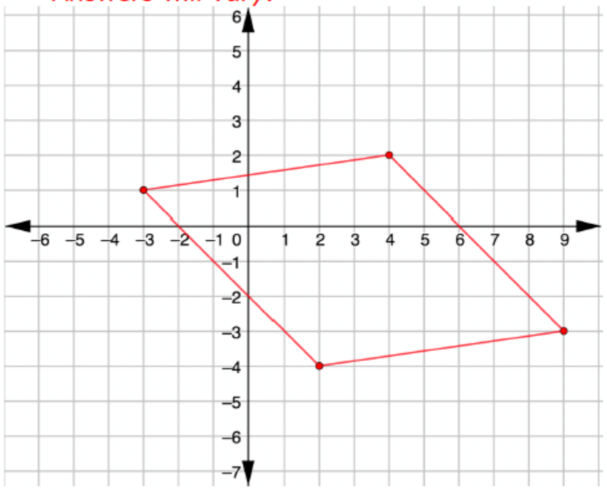
*Consider allowing students to use graph paper or an online graphing utility with question 1 to provide entry points for visual learners.*

*Consider using Instructional Routine, "Take Turns," during question 1. Students work with a partner or a small group. Students will take turns finding the length of sides or the diagonals. Students will be explaining, justifying, and asking clarifying questions within their group.*

*Consider providing students who might be struggling with a table to stay organized, and to have a better visual of the connections between side lengths and slopes.*

# 13.8.4 Guided Practice: Classifying Quadrilaterals (continued)

3. Use the coordinate grid to complete the following.
   a. Draw a quadrilateral for which the side lengths are all equal, but the diagonals are not congruent.
   *Answers will vary.*



   b. Complete the statement.
   This quadrilateral is classified as a ___*rhombus*___.

4. Discuss with your partner how another classmate could use the coordinates of your quadrilateral to verify that your classification is correct. Summarize your discussion.
   *Answers will vary.*
   *They could use the Pythagorean theorem to find the side measures of the quadrilateral to see if all sides measure the same. Then, they could use the Pythagorean theorem to find the length of the diagonals to determine that they are not equal.*

*Consider having students trade their quadrilateral with a partner to clarify the quadrilateral drawn is a rhombus, and to critique their partner's work, if needed.*

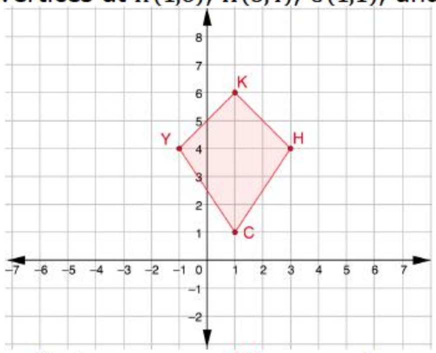
*For students who finish early, consider challenging them to create a quadrilateral, then trade quadrilaterals with a partner to see if their partner can classify their quadrilateral with justifications.*

## 13.8.5 Your Turn!

**Component Overview:** *Students work independently on drawing quadrilaterals to classify them by identifying the properties. Students use previous knowledge from the Exploration and Guided Practice areas to work independently.*

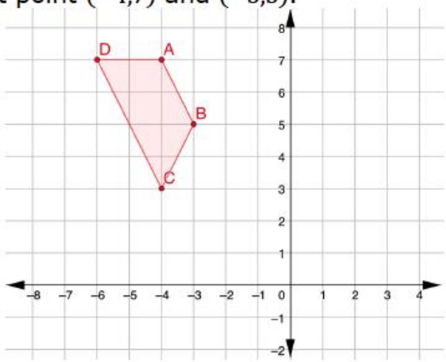
### 13.8.5 Your Turn!

1. $KHCY$ has vertices at $K(1,6)$, $H(3,4)$, $C(1,1)$, and $Y(-1,4)$.




$KHCY$ is classified as a ____*kite*____ because ...
*exactly two pairs of adjacent sides are congruent, but opposite sides are not. Diagonals are perpendicular.*

2. Draw a quadrilateral with one set of parallel sides and vertices at point $(-4,7)$ and $(-3,5)$.




This quadrilateral is classified as a ____*trapezoid*____ because ... *there is one set of parallel sides.*

*Consider providing a table for students who may be struggling to keep their work organized, and to have a better visual on the connections between the side lengths, diagonals, and/or slopes.*

*Consider using guided questions to help students classify the quadrilateral:*

- *What do you notice about the side lengths of the quadrilateral?*
- *What do you notice about the slopes of the side lengths of the quadrilateral?*

*Have students share the quadrilateral they drew with a partner. Each partner should verify that the quadrilateral classified matches the quadrilateral drawn.*

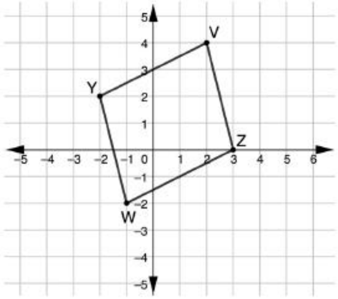
# 13.8.6 Wrap-Up: Error Analysis

***Component Overview:*** *Students work through the Wrap-Up individually to decide if they agree or disagree with the given classification, and to justify why they are using properties and mathematical calculations.*

## 13.8.6 Wrap-Up: Error Analysis

1. $YVZW$ has vertices at $Y(-2,2)$, $V(2,4)$, $Z(3,0)$, and $W(-1,-2)$.

   Aryana graphed quadrilateral $YVZW$ and classified the quadrilateral as a rhombus.




   a. Do you agree with Aryana?
   *No.*

   b. Justify your answer using properties and showing your work.

   $4^2 + 1^2 = WY^2$
   $\sqrt{17} = WY$

   $4^2 + 1^2 = VZ^2$
   $\sqrt{17} = VZ$

   $4^2 + 2^2 = WZ^2$
   $2\sqrt{5} = WZ$

   $4^2 + 2^2 = YV^2$
   $2\sqrt{5} = YV$

   *All four sides do not have equal length.*

*Students' answers on this activity can provide valuable data as to whether students understand the content of this lesson.*